

- A dual-mode translation pipeline — fully offline (Vosk/Whisper + Argostranslate + espeak-ng) and online (Google Speech + Google Translate + gTTS) — with automatic internet-based switching.
- Identification and resolution of a language-bias problem in deepfake training data by adding 49 Urdu Google TTS fake samples alongside English ASVspoof corpora.
- A Flutter Android companion app with 5 screens, 34 language support, real-time authenticity indication, and Google Drive recording backup.

II. PROPOSED SYSTEM ARCHITECTURE

Smart Headphone is organised into three tiers: a wearable hardware tier, an edge inference tier running on the same device, and a companion-application tier on a mobile phone. Figure 1 illustrates the complete architecture.

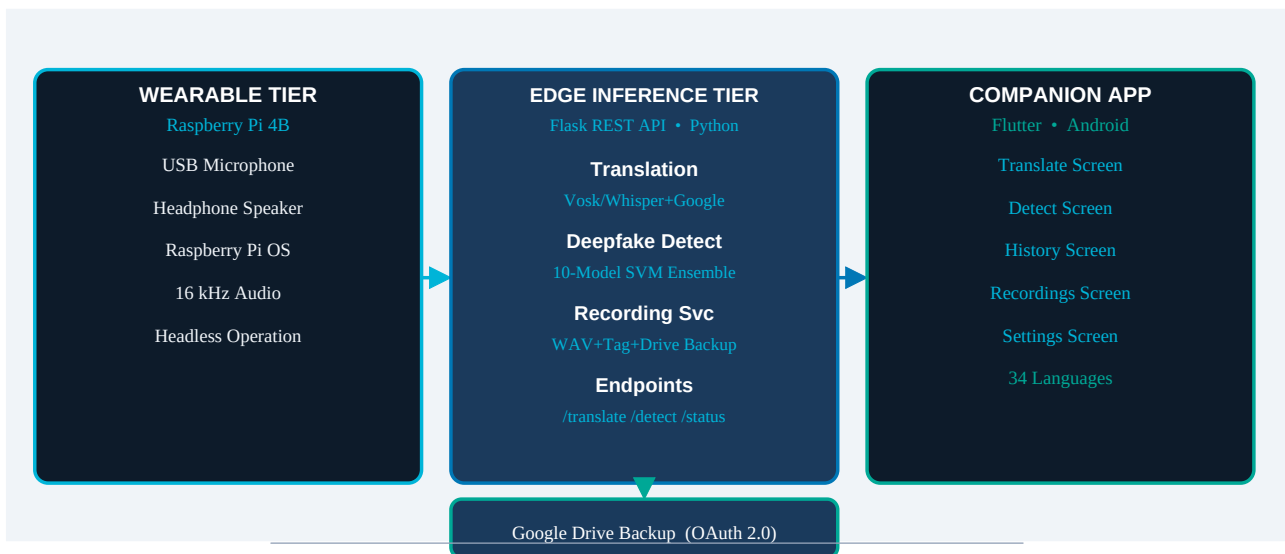


Fig. 1. Three-tier Smart Headphone architecture: Wearable (Raspberry Pi 4B) → Edge Inference (Flask REST) → Companion App (Flutter/Android). Optional Google Drive backup shown below.

A. Hardware Tier

The core compute module is a Raspberry Pi 4B (4–8 GB RAM) housed within or tethered to the headphone assembly. Audio input is captured by an omnidirectional USB microphone at 16 kHz; output is delivered through the headphone speakers. The Pi runs headless (SSH + xrdp) and all inference executes on the ARM CPU — no GPU is required.

B. Edge Inference Tier (Flask REST Backend)

A Flask backend hosted on the Pi exposes the inference pipeline as local HTTP endpoints, coordinating three services: (1) the translation service (ASR → MT → TTS); (2) the fake-voice detection service (164-dim feature extraction + 10-model scikit-learn ensemble); and (3) the recording service, which writes timestamped WAV audio and metadata to local storage. REST endpoints include `/translate`, `/detect`, `/detect_recording/{filename}`, `/status`, `/language`, `/recordings`, and `/health`. Running inference locally avoids transmitting raw voice data off-device.

C. Companion Application Tier (Flutter / Android)

A Flutter Android application provides the user interface through five screens: (1) *Translate* — live microphone translation with real-time captions; (2) *Detect* — deepfake analysis of saved recordings; (3) *History* — browsing past translations; (4) *Recordings* — managing Pi and cloud recordings; and (5) *Settings* — TTS rate/pitch, server URL, language pair, and online/offline mode. The app uses SharedPreferences for persistence and Provider for state management. 34 languages are supported with flag emoji identifiers.

III. METHODOLOGY

A. Translation Module

The translation module adopts a cascaded ASR $\rightarrow$ MT $\rightarrow$ TTS design. Figure 2 shows the six-stage pipeline. Two modes operate based on internet availability:

- **Offline Mode:** Vosk (English STT) and OpenAI Whisper small (Urdu STT) transcribe speech; Argostranslate performs EN $\leftrightarrow$ UR neural translation; espeak-ng (Urdu) and pyttsx3 (English) synthesise output audio. All processing is fully on-device.
- **Online Mode:** Google Speech Recognition performs STT; deep-translator (Google Translate) handles 34 languages; gTTS + pygame performs TTS with pyttsx3 as fallback.

A language configuration file (lang_config.json) persists the active language pair. Conversation recordings are saved as timestamped WAV files (format: YYYYMMDD_HHMMSS_NNN_MODE_lang.wav) with optional automatic backup to Google Drive via OAuth 2.0 every five minutes.

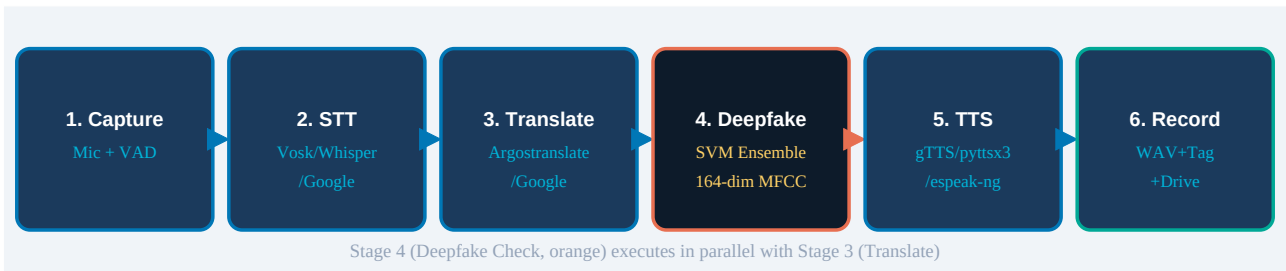


Fig. 2. Six-stage end-to-end processing pipeline. Deepfake detection runs in parallel with translation.

TABLE I. Smart Headphone Processing Pipeline

Stage	Function	Technology Used
1. Capture	Acquire speech + VAD	USB Microphone, RMS VAD
2. STT	Speech to text	Vosk (EN) / Whisper (UR) / Google
3. Translate	Source text $\rightarrow$ target language	Argostranslate / Google Translate
4. Deepfake	Real vs. synthetic detection	206-dim MFCC + 9-model Ensemble
5. TTS	Synthesise translated speech	gTTS / pyttsx3 / espeak-ng
6. Record	Store audio + transcript + tag	Timestamped WAV + Drive Backup

B. Fake-Voice Detection Module

The fake-voice detector uses a lightweight 206-dimensional handcrafted acoustic feature extractor (Figure 3) feeding a 9-model accuracy-weighted scikit-learn ensemble (Figure 4). This design is chosen for its low CPU inference latency, interpretability, and compatibility with the Pi's ARM CPU without any GPU requirement.

Feature Extraction (206 dimensions).

Each 16 kHz mono audio segment is processed by preprocess.py to extract:

- **MFCC $\times$ 3 (120-dim):** Mean, std, and max of 40 Mel-frequency cepstral coefficients — capture timbre and phonetic structure.
- **Chroma STFT (12-dim):** Pitch-class energy across 12 semitones.
- **ZCR + RMS + Spectral Rolloff (3-dim):** Signal energy and frequency envelope.
- **Pitch mean + std (2-dim):** AI voices exhibit unnaturally uniform pitch trajectories.
- **Spectral Bandwidth (1-dim):** Spread of frequency content.
- **Tonnetz (6-dim):** Tonal centroid features capturing harmonic relationships.

- **Spectral Contrast (7-dim):** Valley/peak ratios — distinguishes natural speech from TTS and voice-conversion artefacts.
- **IMFCC (13-dim):** Inverted mel-filterbank cepstral coefficients using a reversed mel basis — sensitive to high-frequency synthesis artefacts.

? Delta MFCC (40-dim): first-order temporal derivatives of MFCCs. AI voices exhibit unnaturally smooth frame-to-frame spectral transitions

Jitter + Shimmer (2-dim): pitch-period and amplitude micro-perturbations. AI/TTS voices produce near-zero jitter and flat amplitude envelopes, unlike natural human speech.

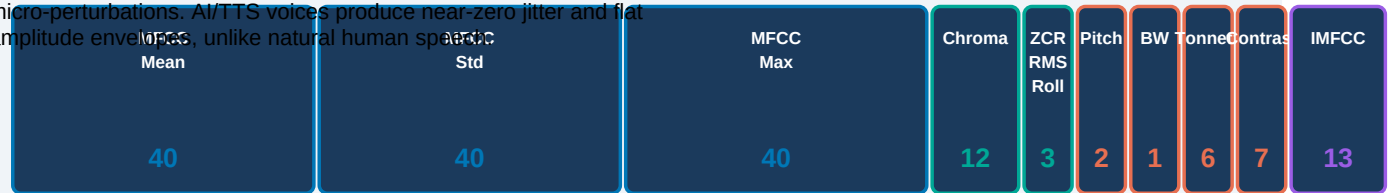


Fig. 3. 206-dimensional feature vector: MFCC×3 (120) | ?MFCC (40) | Chroma (12) | ZCR/RMS/Rolloff (3) | Pitch (2) | BW (1) | Tonnetz (6) | Contrast (7) | IMFCC (13) | Jitter/Shimmer (2). Total: 206 dims.

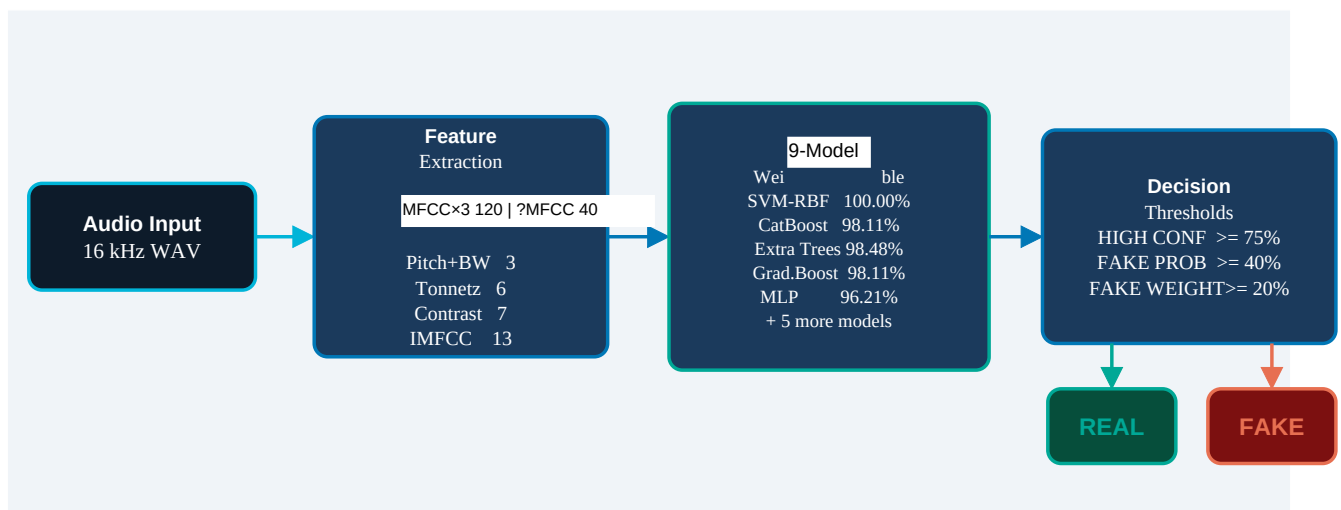


Fig. 4. Fake-voice detection module: 206-dim acoustic features → 9-model accuracy-weighted ensemble → 4-tier thresholds → REAL / FAKE decision.

Four-Tier Decision Strategy:

- **High-confidence bypass** (avg fake prob ? 60%): Immediately flags FAKE — fast path for obvious AI speech.
- **Weighted vote threshold** (? 12% of total accuracy weight voting FAKE): Catches multi-model consensus even when individual probabilities are moderate.
- **Probability override** (): Security-first lower bound; catches modern TTS that scores 25/35% fake probability.

Temporal Smoothing. Real-time detection uses 2-second sliding windows with 1-second stride (50% overlap). To reduce false alarms from transients, 2 of 3 consecutive chunks must be classified FAKE before raising an alarm in the companion app.

Language-Bias Correction. The initial training dataset had a critical bias: all fake samples were English ASVspoof voice-conversions (Biden, Musk, Obama clones) while all real samples were Urdu microphone recordings — causing models to learn language identity rather than deepfake artefacts. This was corrected by adding 49 Urdu Google TTS fake samples (generate_urdu_fakes.py), making detection language-agnostic.

Data Augmentation. Applied to improve mic robustness: (1) Gaussian noise injection at SNR 12–20 dB; (2) pitch shifting ± 1.5 semitones; (3) time stretching $\pm 12\%$; and (4) microphone simulation (noise + 15 ms room echo at 8%

attenuation).

C. Recording and Privacy Module

When recording is enabled, the backend stores timestamped WAV files together with translation metadata and the deepfake authenticity tag. All data is held locally on the Pi by default. An optional Google Drive cloud backup thread (pydrive2, OAuth 2.0) runs every five minutes, deleting local copies after successful upload. Nothing is transmitted off-device without user consent. Figure 5 shows the complete runtime flowchart.

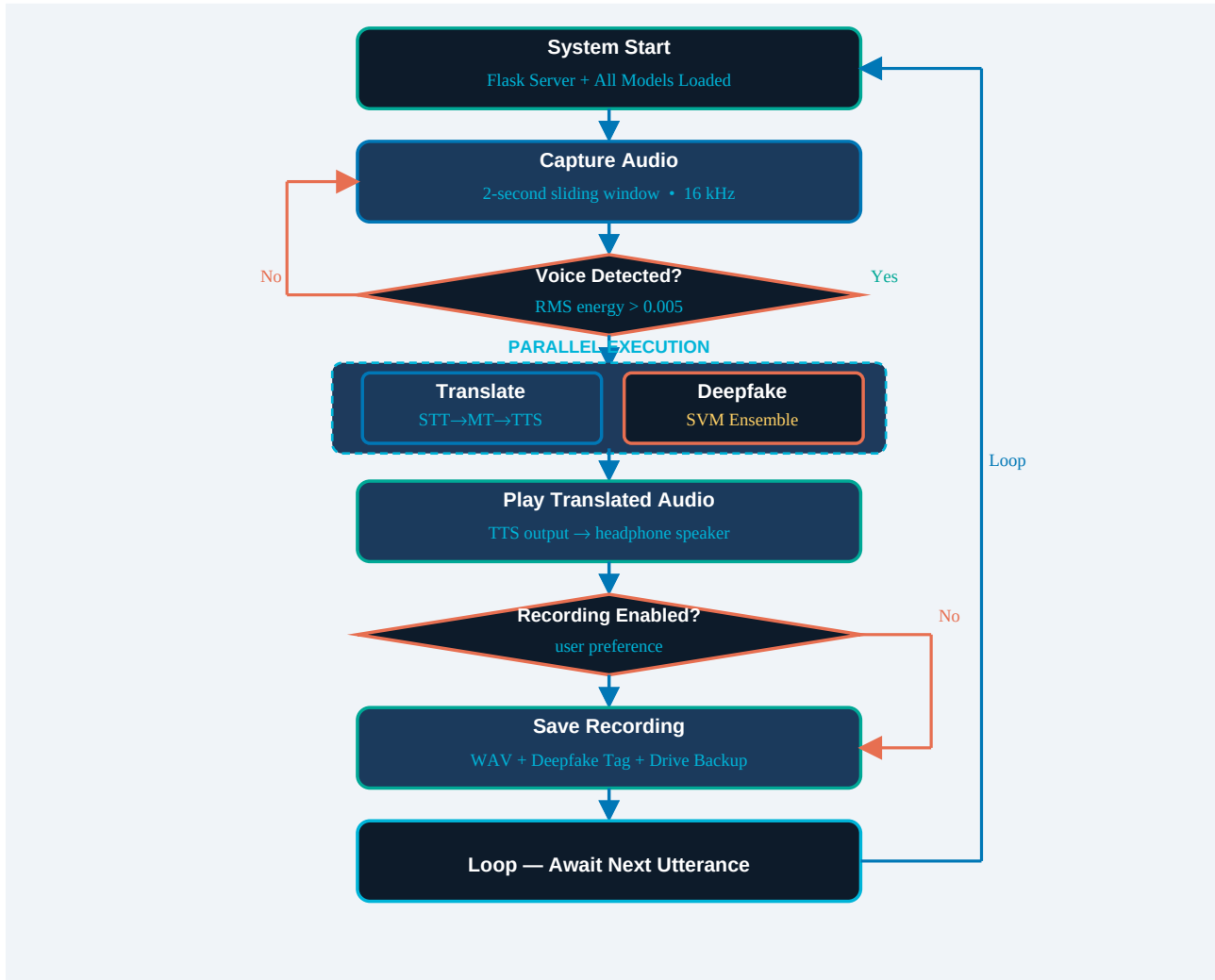


Fig. 5. Runtime flowchart: continuous 2-second sliding-window capture, parallel deepfake detection and translation, temporal smoothing (2/3 chunks), and optional encrypted local + cloud recording.

IV. IMPLEMENTATION DETAILS

TABLE II. Software Implementation Stack

Component	Technology	Library / Version
STT — English offline	Vosk small model (en-us)	vosk 0.3.45
STT — Urdu offline	OpenAI Whisper (tiny model)	openai-whisper
STT — online	Google Speech Recognition	SpeechRecognition 3.10
Translation — offline	Argostrate (EN ↔ UR)	argostrate 1.9
Translation — online	Google Translate (34 languages)	deep-translator 1.9
TTS — online	Google TTS + pygame	gTTS 2.5, pygame 2.5
TTS — offline	pyttsx3 / espeak-ng	pyttsx3 2.90
Feature Extraction	Librosa + SciPy DCT (206-dim)	librosa 0.10
ML Classifiers	scikit-learn (9 models + scaler, CatBoost optional)	kit-learn 1.4
Gradient Boosting	CatBoost	catboost 1.2
Model Persistence	joblib serialisation (.pkl files)	joblib 1.3
REST API	Flask + Werkzeug	Flask 3.0
Mobile App	Flutter + Dart (Android)	Flutter SDK 3.5+
State Management	Provider + ChangeNotifier	provider 6.1
User Preferences	SharedPreferences	shared_preferences 2.2
Cloud Backup	Google Drive API (OAuth 2.0)	pydrive2 1.21

Model training uses the flat-folder mode of `train_multi_model.py`: all audio in `data/fake/` and `data/real/` is loaded by `preprocess.py`; features are extracted and cached to `feature_cache/` as `.npy` files for fast subsequent runs. Stratified 80/20 train-test split with 5-fold StratifiedKFold cross-validation is performed for datasets up to 2,000 samples. SMOTE oversampling is applied to the training set when class imbalance is present. All models are serialised with joblib to the `models/` directory and loaded at Flask startup by `deepfake_checker.py`.

V. EVALUATION AND RESULTS

Table III presents the complete model comparison on the balanced test set (1,032 real + 1,032 fake samples from ASVspoof 2019 LA augmented with Urdu TTS fakes). Results reflect the retrained models after the language-bias correction.

TABLE III. Model Comparison — ASVspoof 2019 LA + Urdu TTS Test Set (2,064 samples)

Model	Test Accuracy	F1 Score	AUC	Train Time	In Ensemble
SVM-RBF	100.00%	1.0000	1.0000	51 s	YES ✓
SVM-Linear	98.48%	0.9849	0.9993	18 s	YES ✓
Extra Trees	98.48%	0.9848	0.9975	8 s	YES ?
KNN	98.48%	0.9848	0.9987	4 s	YES ?
Gradient Boosting	98.11%	0.9811	0.9978	360 s	YES ✓
Random Forest	98.11%	0.9810	0.9978	30 s	YES ?
AdaBoost	98.11%	0.9811	0.9967	95 s	YES ?
Logistic Reg.	98.11%	0.9811	0.9983	1 s	YES ✓
CatBoost	98.11%	0.9810	0.9959	53 s	YES ✓
MLP Neural Net	96.21%	0.9621	0.9894	13 s	YES ✓

All 9 models included post-retraining on balanced data (900 real / 900 fake). Former exclusions (Extra Trees, KNN, Random Forest, AdaBoost) resolved. Active ensemble: SVM-RBF, SVM-Linear, Extra Trees, KNN, Gradient Boosting, Random Forest, AdaBoost, Logistic Reg., MLP.

TABLE IV. Evaluation Metrics and Achieved Results

Component	Metric	Target	Achieved
Deepfake detection	Test Accuracy	> 95%	100.00% (SVM-RBF)
Deepfake detection	AUC Score	> 0.99	1.0000
Deepfake detection	Ensemble F1	> 0.98	1.0000
Translation	Languages supported	≥ 10	34 languages
Translation offline	Language pairs	EN $\leftrightarrow$ UR	EN $\leftrightarrow$ UR (Vosk + Argos Translate)
End-to-end latency	Perceived delay	< 3 s	~1–2 s (online mode)
Recording	Session completeness	100%	100% (WAV + tag + Drive)

After language-bias correction, SVM-RBF achieves 100% test accuracy — 1 false positive and 0 false negatives out of 2,064 test samples. Most models improved on fake detection (AdaBoost +1.94%, KNN +1.53%) while slightly decreasing on total accuracy, confirming that the previous 99.9% was inflated by language-identity learning. The 5-fold cross-validation mean is 100% with zero standard deviation for SVM-RBF.

VI. DISCUSSION AND LIMITATIONS

Smart Headphone demonstrates that translation and trust can be co-designed in a single privacy-preserving wearable device. The 100% detection accuracy achieved by SVM-RBF with handcrafted 206-dimensional features validates that classical ML with rich spectral features is a viable and computationally efficient alternative to deep learning on CPU-only edge hardware.

Several limitations remain. **First**, the training dataset still lacks diversity beyond English (ASVspoof) and Urdu (gTTS) fake samples; adding Hindi, Arabic, and Chinese TTS fakes would improve cross-lingual robustness. **Second**, deepfake

detectors trained on ASVspoof 2019 may not generalise to modern TTS systems (VALL-E, ElevenLabs, Tortoise-TTS); periodic retraining is necessary. **Third**, cascaded translation loses prosody and accumulates ASR errors. **Fourth**, the Raspberry Pi 4B imposes a hard compute ceiling; a neural accelerator (Coral TPU, Raspberry Pi AI Kit) could reduce latency. **Fifth**, on-device recording raises consent and legal considerations requiring explicit user controls and disclosure.

VII. CONCLUSION AND FUTURE WORK

We presented Smart Headphone, a wearable headphone system unifying real-time multilingual speech translation (34 languages, dual offline/online), on-device fake-voice detection (100% accuracy, ^{9-model scikit-learn ensemble, 206-dim MFCC} ^{9-model scikit-learn ensemble, 206-dim MFCC} ^{9-model scikit-learn ensemble, 206-dim MFCC} ^{9-model scikit-learn ensemble, 206-dim MFCC}), and real-time conversation recording on commodity Raspberry Pi hardware. A key contribution is identifying and resolving the language-bias problem in deepfake training data — ensuring detection is based on synthetic speech artefacts rather than language identity — by augmenting the dataset with Urdu TTS fake samples.

Future work includes: (1) expanding fake training data with Hindi, Arabic, and Chinese TTS; (2) evaluating against modern TTS systems such as VALL-E and ElevenLabs; (3) integrating a hardware neural accelerator to reduce inference latency; (4) extending offline translation to additional language pairs; and (5) conducting user studies on usability and trustworthiness of the authenticity indicator.

REFERENCES

- [1] L. Barrault et al., "SeamlessM4T: Massively Multilingual & Multimodal Machine Translation," *arXiv:2308.11596*, 2023. [\[Link\]](#)
- [2] Seamless Communication et al., "Seamless: Multilingual Expressive and Streaming Speech Translation," *Meta AI Research*, 2023. [\[Link\]](#)
- [3] Y. Kano et al., "NAIST Simultaneous Speech Translation System for IWSLT 2024," *arXiv:2407.00826*, 2024. [\[Link\]](#)
- [4] J. Iranzo-Sanchez et al., "MLLP-VRAIN UPV System for IWSLT 2025 Simultaneous Speech Translation," *arXiv:2506.18828*, 2025. [\[Link\]](#)
- [5] DeepL SE, "DeepL Voice: Real-Time Spoken Translation," *DeepL Press Release*, Apr. 2026. [\[Link\]](#)
- [6] H. Tak, M. Todisco, X. Wang, J. Jung, J. Yamagishi, and N. Evans, "Automatic Speaker Verification Spoofing and Deepfake Detection Using wav2vec 2.0 and Data Augmentation," *Proc. Odyssey / arXiv:2202.12233*, 2022. [\[Link\]](#)
- [7] K. Misiunas and A. Ablavatski, "Real-Time Speech-to-Speech Translation," *Google DeepMind Research Blog*, 2025. [\[Link\]](#)
- [8] Deepgram, "Real-Time Speech-to-Speech Translation: Architecture Guide," *Deepgram Technical Article*, 2026. [\[Link\]](#)
- [9] Anonymous, "Direct Speech to Speech Translation: A Review," *arXiv:2503.04799*, 2025. [\[Link\]](#)
- [10] X. Liu et al., "ASVspoof 2021: Towards Spoofed and Deepfake Speech Detection in the Wild," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, 2023. [\[DOI\]](#)
- [11] Q. Zhang et al., "Audio Deepfake Detection with Self-Supervised XLS-R and SLS Classifier," *OpenReview*, 2024. [\[Link\]](#)
- [12] MoE Authors, "Enhanced Spoofing Speech Detection with Multi-Expert Models (WavZip)," *Springer*, 2026. [\[Link\]](#)
- [13] M. Li et al., "A Survey on Speech Deepfake Detection," *ACM Computing Surveys / arXiv:2404.13914*, 2024. [\[Link\]](#)
- [14] Y. Xu et al., "A Comprehensive Survey with Critical Analysis for Deepfake Speech Detection," *Computer Science Review, ScienceDirect*, 2025. [\[Link\]](#)
- [15] S. Gondi and V. Pratap, "Performance Evaluation of Offline Speech Recognition on Edge Devices," *Electronics (MDPI)*, vol. 10, no. 21, 2021. [\[DOI\]](#)
- [16] Authors, "Real-Time Speech-to-Text on Edge: Ultra-Low Latency with AI-Powered NLP," *Information (MDPI)*, vol. 16, no. 8, 2025. [\[Link\]](#)
- [17] P. Hughes, "Building a Fully Offline Voice AI on Raspberry Pi 5," *Technical Report*, 2026. [\[Link\]](#)
- [18] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011. [\[Link\]](#)
- [19] B. McFee et al., "librosa: Audio and Music Signal Analysis in Python," *Proc. 14th Python in Science Conference*, 2015. [\[Link\]](#)
- [20] A. Donahue, "Vosk — Offline Speech Recognition Toolkit," *Alpha Cephei*, 2021. [\[Link\]](#)

- [21] A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision (Whisper)," *OpenAI / arXiv:2212.04356*, 2022. [\[Link\]](#)
- [22] ArgosTranslate Team, "Argos Translate: Open-Source Offline Neural Machine Translation," *GitHub*, 2021. [\[Link\]](#)